CPT202 2024/2025 Semester 2
Software Requirements Report for Software Engineering Group Project

Project C: An Online Meeting Booking System

Group Number: 33

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Understanding of the Scrum framework

Scrum is an Agile framework for managing complex projects through iterative Sprints (2-4 weeks). Roles include the Product Owner (prioritizes PBIs), Scrum Master (facilitates), and Developers. PBIs (Product Backlog Items) are user-centric requirements, like user stories with acceptance criteria, defining deliverables. The team selects PBIs for each Sprint, breaking them into tasks. Daily Scrums track progress. By the end of Sprint, a working increment is delivered. PBIs ensure alignment with user needs, prioritized in the Product Backlog, guiding development and enabling adaptability through feedback. They translate stakeholders' needs into actionable work, ensuring transparency and continuous improvement.

(96 words)

(PBIs See Page 3)

PBIs

PBI01: User View Information about the Meeting Room

- **User Story**: As a user, I want to be able to view detailed information about the meeting room (room capacity, available time, facilities, etc.) in order to choose the appropriate meeting room based on actual needs.
- **Acceptance Criteria**:
  1. **Accessing the room detail page**
     - **Given** that the user has logged in and is on the meeting room list page.
     - **When** the user clicks on the "View Details" button corresponding to a conference room in the list.
     - **Then** the system should open the room detail page in the new tab and display all relevant information about the room.
  2. **Displaying detailed room information**
     - **Given** the room detail page has loaded.
     - **When** the page is displayed.
     - **Then** it must display details including the picture of the room, room No. and location, capacity, layout, available time slots, IT infrastructures, and additional descriptions.
  3. **Loading Indicator**
     - **Given** the data load is delayed due to slow network or large data volume.
     - **When** the user clicks on the "View Details" button.
     - **Then** the system should show a loading animation until all information is fully loaded.
  4. **Responsive Design**
     - **Given** the user accesses the room detail page on various devices (desktop, tablet, mobile)
     - **When** the screen size and pixel density (PPI) changes
     - **Then** the page layout should automatically adjust to ensure all information remains clear and accessible.
  5. **Accessibility Requirements**
     - **Given** the user is using a keyboard-only interface or screen-reading software
     - **When** the user reviews the room details
     - **Then** all information and controls should be correctly accessible and operable without requiring a mouse.

PBI02: User make meeting reservations

- **User Story**: As a user, I want to book a meeting room online so that I can efficiently schedule meetings and secure the required room resources.
- **Acceptance Criteria**:
  1. **Initiating the Booking Process**
     - **Given** the user is logged in and on the room details page.
     - **When** the user clicks on the available time slots [fig. 1] in the timetable which specifies the beginning and ending time of the reservation.
     - **Then** the system should display the booking form pre-filled with the corresponding time slot information.
  2. **Entering Booking Information**
     - **Given** the user is filling in the booking form (meeting topic, number of attendees, etc.) [fig. 2]
     - **When** the user enters information
     - **Then** the input fields should display the user's entries and error message (if exist) in real time.
  3. **Submitting the Booking Request**
     - **Given** the user has filled in all required fields and selected available time slot
     - **When** the user clicks the "Book" button
     - **Then** the system should verify the room's availability and, if valid, submit the booking request and display a "Booking Successful" message, else display the time conflict message.
  4. **Performance Requirements**
     - **Given** the booking information is complete
     - **When** the user clicks the "Book" button
     - **Then** the system should respond within 2 seconds and validate inputs within 1 second.
  5. **Responsive Design**
     - **Given** the user is booking a meeting on various devices (desktop, tablet, mobile)
     - **When** the screen size and pixel density (PPI) changes
     - **Then** the booking form should adjust its layout to remain user-friendly.
  6. **Accessibility Requirements**
     - **Given** the user is utilizing assistive tools
     - **When** the user interacts with the booking form
     - **Then** all controls (input fields, buttons, error messages) must be accessible and **operable** via keyboard, touch and other assistive technology.

PBI03: User View Meeting Booking History

- **User Story**: As a user, I want to view all my meeting booking records—including completed, ongoing and upcoming meetings, so that I can manage my schedule effectively.
- **Acceptance Criteria**:
  1. **Accessing the Booking History Page**
     - **Given** the user is logged in
     - **When** the user clicks on the "Booking History" section
     - **Then** the system should load and display a list of all booking records.
  2. **Displaying Booking History with Category**
     - **Given** that the user has any reservation for the corresponding category.
     - **When** the page loads
     - **Then** the system should clearly label the meetings with different statuses (completed/ ongoing/ upcoming) and show detailed information (time, room location, etc.) in every row.
  3. **Display reservation history in pagination**
     - **Given** the user has reservation records exceeding one page capacity
     - **When** the user clicks the page No. button or inputs the page No. to jump
     - **Then** the system needs to load the corresponding page within 2 seconds.
  4. **Filtering and Sorting Options**
     - **Given** the user wants to filter booking records by date or status
     - **When** the user applies the filtering or sorting options
     - **Then** the list should be dynamically updated to meet the applied criteria.
  5. **Loading and no result indication**
     - **Given** the user wants to filter booking records by date or status
     - **When** the user applies the filtering or sorting options
     - **Then** during the loading time, the system should show a loading animation until the information is fully loaded. And if the result is empty, system should show “no result” message.
  6. **Performance Requirements**
     - **Given** the user wants to filter the reservation records
     - **When** the user clicks the “filter” or “sort” button
     - **Then** the system should display the results within 1 second.
  7. **Accessibility Requirements**
     - **Given** the user is navigating with assistive technologies
     - **When** browsing the booking history
     - **Then** all records and interactive elements should be accessible via screen readers and keyboard navigation.

PBI04: User Cancel Upcoming Meeting Booking

- **User Story**: As a user, I want to cancel a booked meeting so that I can free up meeting room resources when my plans change.
- **Acceptance Criteria**:
  1. **Initiating the Cancellation Process**
     - **Given** that the user has an upcoming meeting booking (at least 5 minutes before the meeting starts) in their history
     - **When** the user clicks the "Cancel Booking" button for that record in the booking history page.
     - **Then** the system should display a confirmation dialog asking the user to confirm the cancellation or not.
  2. **Confirming the Cancellation**
     - **Given** the cancellation confirmation dialog is displayed
     - **When** the user clicks the "Confirm Cancellation" button
     - **Then** the system should cancel the booking, update the booking status, and display a "Cancellation Successful" message.
  3. **Preventing Cancellation of Completed Meetings**
     - **Given** the booking corresponds to a meeting that has already started
     - **When** the user attempts to cancel it
     - **Then** the system should display an error message "Cannot cancel a completed/ ongoing meeting" and disallow the cancellation.
  4. **Network Error Handling**
     - **Given** that a network error occurs during the cancellation process
     - **When** the cancellation request fails
     - **Then** the system should notify the user with an error message and ensure that the booking status remains unchanged
  5. **Responsive Design**
     - **Given** the user is using different devices to cancel a booking
     - **When** the screen size changes
     - **Then** the cancellation interface (buttons, messages) should adjust accordingly.
  6. **Notify meeting members of the cancelation of the meeting room.**
     - **Given** the user submitted the cancellation.
     - **When** the meeting room has confirmed to be cancelled successfully
     - **Then** the system should send reminder emails to the meeting members with the information that the corresponding meeting is cancelled.

PBI05: User Modify Meeting Room

- **User Story**: As a user, I want to modify the meeting room for an upcoming booking so that I can switch to a room that better suits my meeting needs.
- **Acceptance Criteria**:
  1. **Entering Modification Mode**
     - **Given** the user selects an upcoming meeting booking (at least 5 minutes before the meeting starts) from the booking history
     - **When** the user clicks on the "Change Meeting Room" button
     - **Then** the system should display a modification interface showing the current room details and a list of available alternative rooms (available for the corresponding time slot)
  2. **Selecting a New Meeting Room**
     - **Given** the modification interface displays a list of available meeting rooms
     - **When** the user clicks on a new meeting room
     - **Then** the selected room should be highlighted, and its details should be updated in real time in the booking information.
  3. **Submitting the Modification Request**
     - **Given** the new meeting room has been chosen by the user and passed the availability check
     - **When** the user clicks the "Save Changes" button
     - **Then** the system should update the booking details, display a "Meeting Room Change Successful" message, and reflect the new room information in the booking record.
  4. **Handling Unavailable Room**
     - **Given** that the new meeting room becomes unavailable during submission
     - **When** the user clicks the "Save Changes" button
     - **Then** the system should display an error message "The selected room is not available for the chosen time slot" and prevent the change.
  5. **Cancelling the Modification**
     - **Given** the user is in the process of modifying the meeting room
     - **When** the user clicks the "Cancel" button
     - **Then** the system should exit the modification mode and restore the original meeting room details.
  6. **Notify meeting members of the modification to the meeting room.**
     - **Given** the user submitted the modification to the meeting room.
     - **When** the meeting room has confirmed to be modified successfully
     - **Then** the system should send reminder emails to the meeting members.

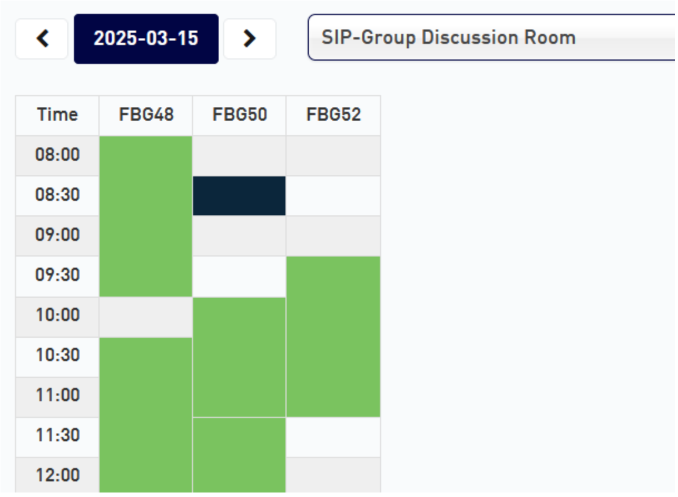

Fig. 1: Example of time slots.

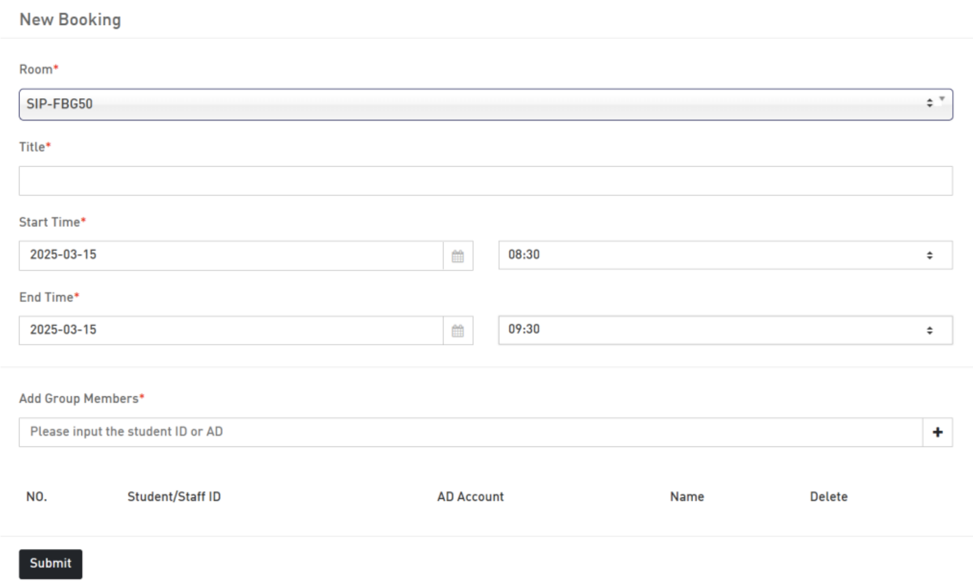

Fig. 2: Example of a booking form.

We can include images to make the PBI clearer

In fact, it is a good practice to include images to help understanding.

In your assignment, try to use images appropriately.

According to "Group Project Preparation.pptx" Page 26, we can include images to make PBI clearer.

Collaboration and Reflection

Peer Review and Feedback

Our group used online documents for PBI allocation and development and then conducted a comprehensive peer review. Although the result generally followed Scrum principles, it also identified some common issues. Below are **three typical problems and my constructive feedback on them**.

The first problem is that the 'Then' statement in the Acceptance Criteria lacks clarity. For example, in the PBI “Administrator Rejects User's Meeting Booking” it is sentenced that “Then … and notify the user about the result.” To provide a more specific description, it is necessary to **specify what kind of notifications (such as emails or system messages)** users will receive after the status update.

The second problem focuses on the incompleteness of error handling. For example, in the case of "Login with incorrect password", although feedback displaying error messages is designed, it does not involve the subsequent impact on system security, **such as "entering a verification code after three consecutive password errors"**. In some scenarios where operations fail, many acceptance criteria forget to cancel the changes caused by previous operations, where the **consistency of operations should be maintained**.

The third problem is that some of the PBIs did not follow the Independent principle of “**INVEST**”. For example, PBI3 “Administrator Delete User’s Account” has strong dependency with other PBIs related with user’s operation. While the PBI3 **can be split** into PBI3.1“Administrator Delete User’s Account” and PBI3.2 “When a user is deleted, how should all related meeting bookings be handled”.

(245 words)

Reflection on Learning

Through the coursework and peer review process, reviewing and developing PBIs helped me recognize the necessity of well-defined acceptance criteria, comprehensive error-handling, and properly decoupled tasks. For instance, PBI01’s loading animation and accessibility requirements highlighted user-centric design, while PBI04’s error-handling emphasized robust system resilience. Peer feedback pushed me to clarify ambiguous criteria (e.g., specifying email notifications in PBI04) and decouple tasks to align with the “INVEST” principle. This experience enhanced my attention to detail and problem-solving skills, reinforcing the value of iterative improvement in Agile development.

(86 words)